INCH-POUND

MIL-DTL-16377/78
18 December 2008

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; T8 FLUORESCENT, GENERAL LIGHTING,17 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 331.1T8, 331.2T8, 347.2T8, AND 347.3T8

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions in inches. For tolerances, other than specified herein, see 3.5.2 of MIL-DTL-16377.

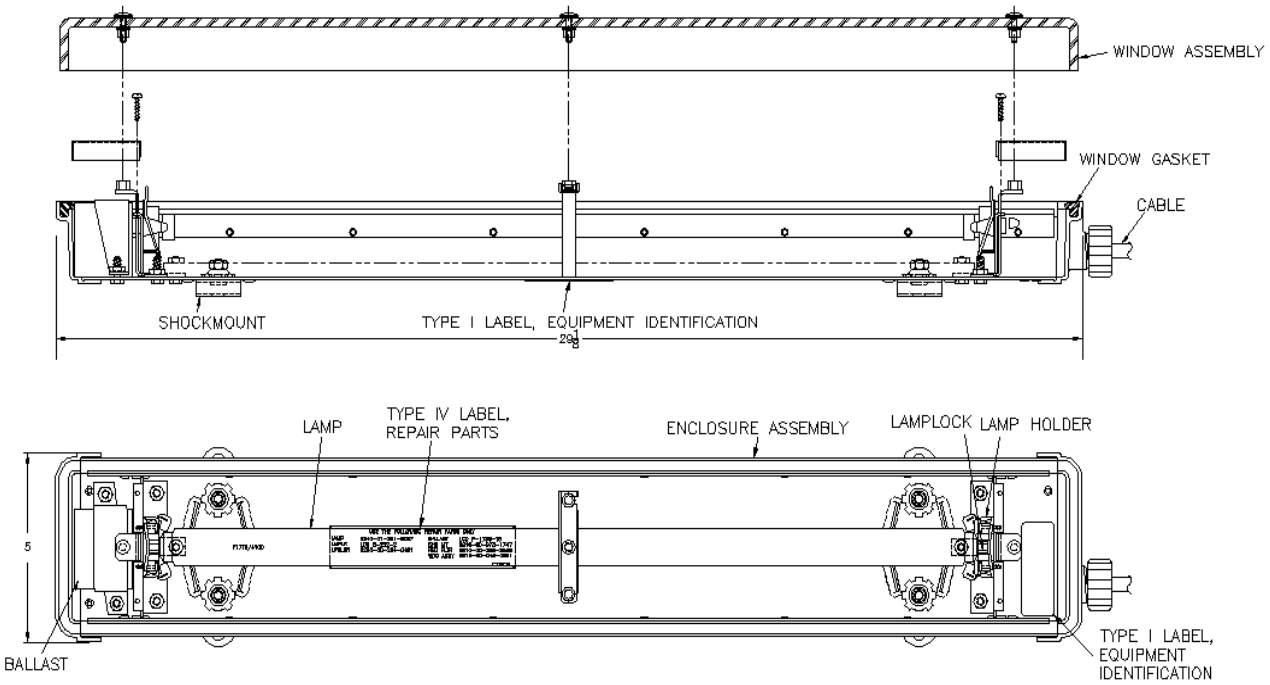


FIGURE 1. Typical view of symbols 331.1T8 and 347.2T8, symbols 331.2T8, and 347.3T8 use identical construction with flush mounting flange not shown.

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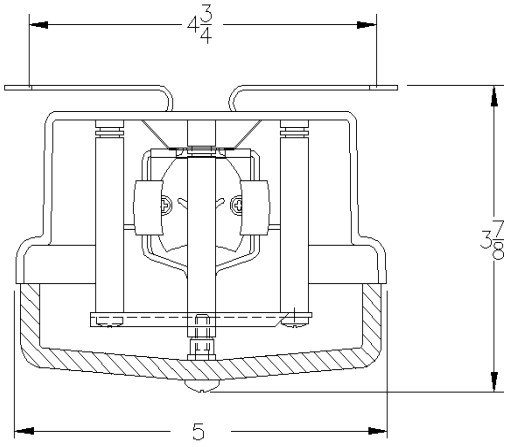
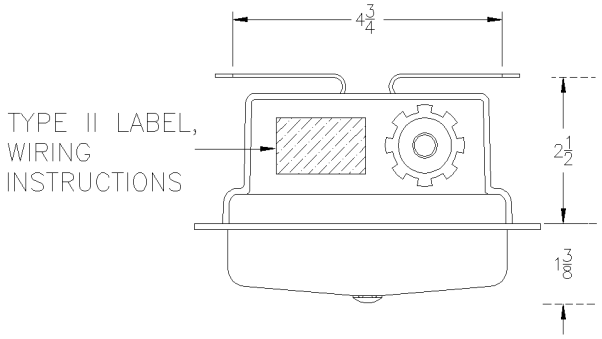


FIGURE 2. Typical section for symbols 331.1T8 and 347.2T8.



NOTE:

1. Type II label required for all fixtures.

FIGURE 3. Typical section for symbols 331.2T8 and 347.3T8.

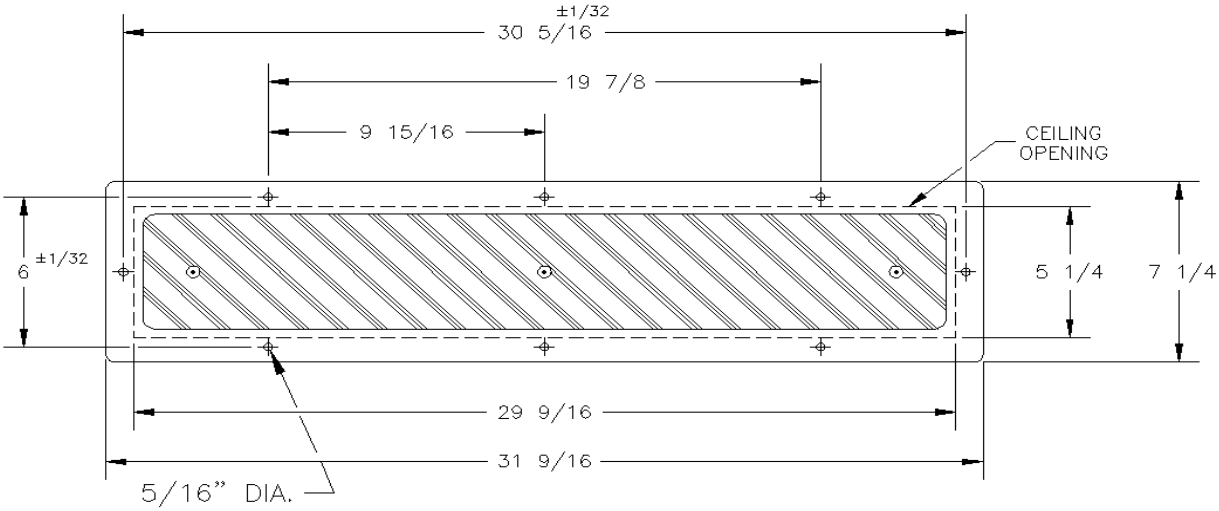


FIGURE 4. Flush mounting dimension for symbols 331.2T8 and 347.3T8.

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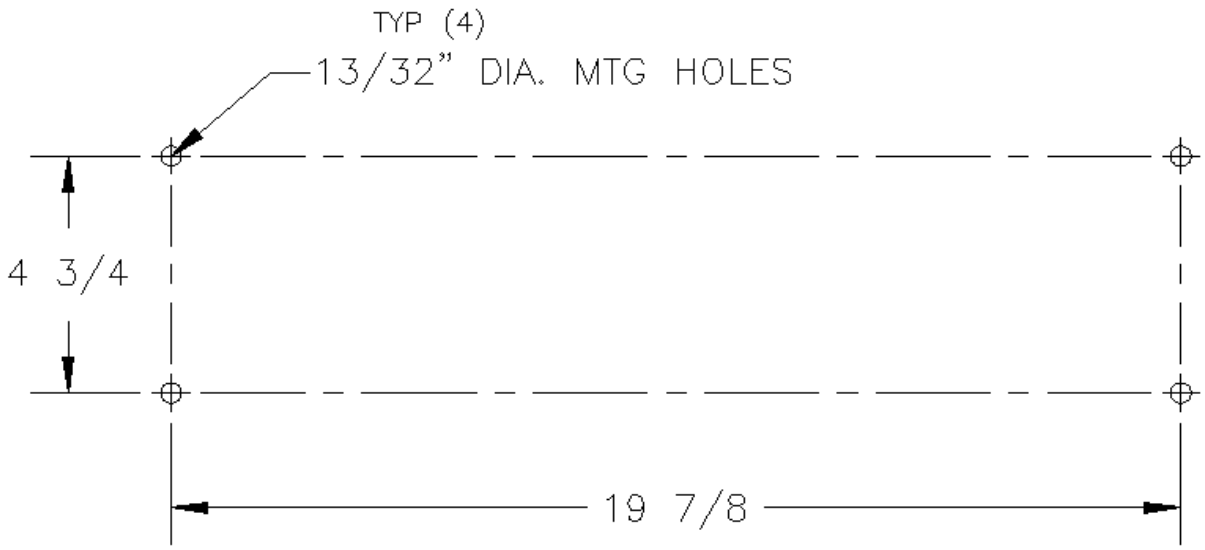


FIGURE 5. Standard mounting dimensions utilizing shock mounts for symbols 331.1T8 and 347.2T8.

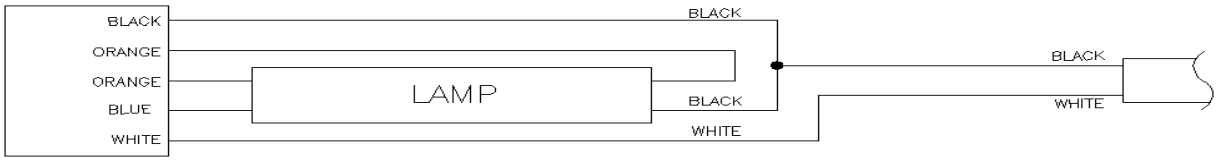


FIGURE 6. Typical wiring diagram.

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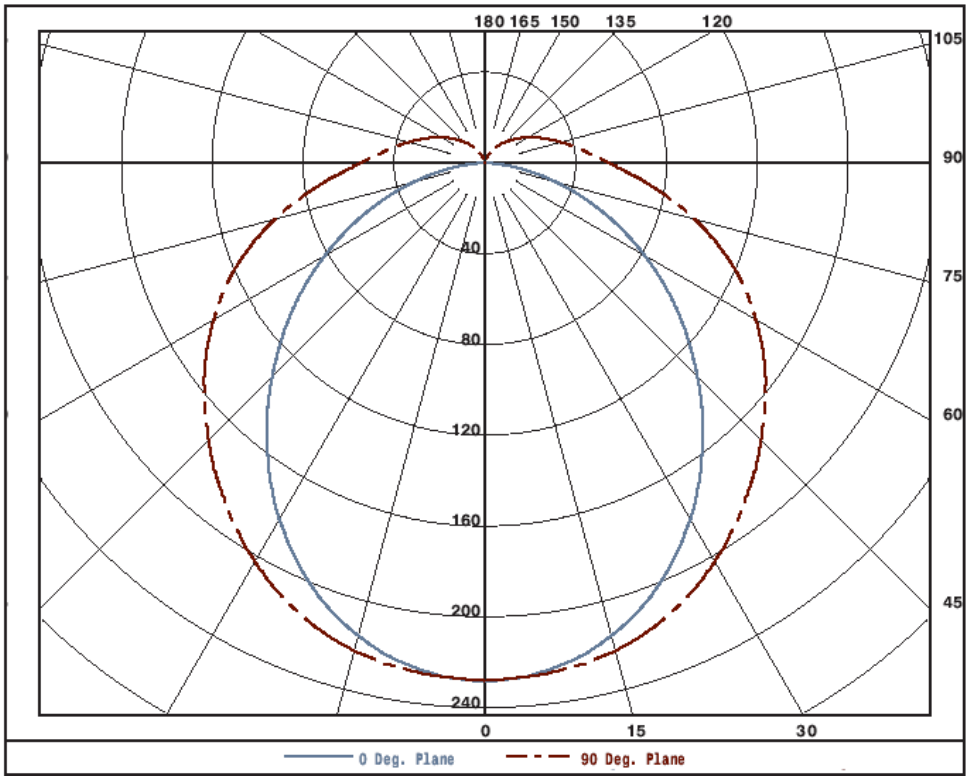


FIGURE 7. Candela distribution curves, white diffusing window, symbols 331.1T8 and 331.2T8.

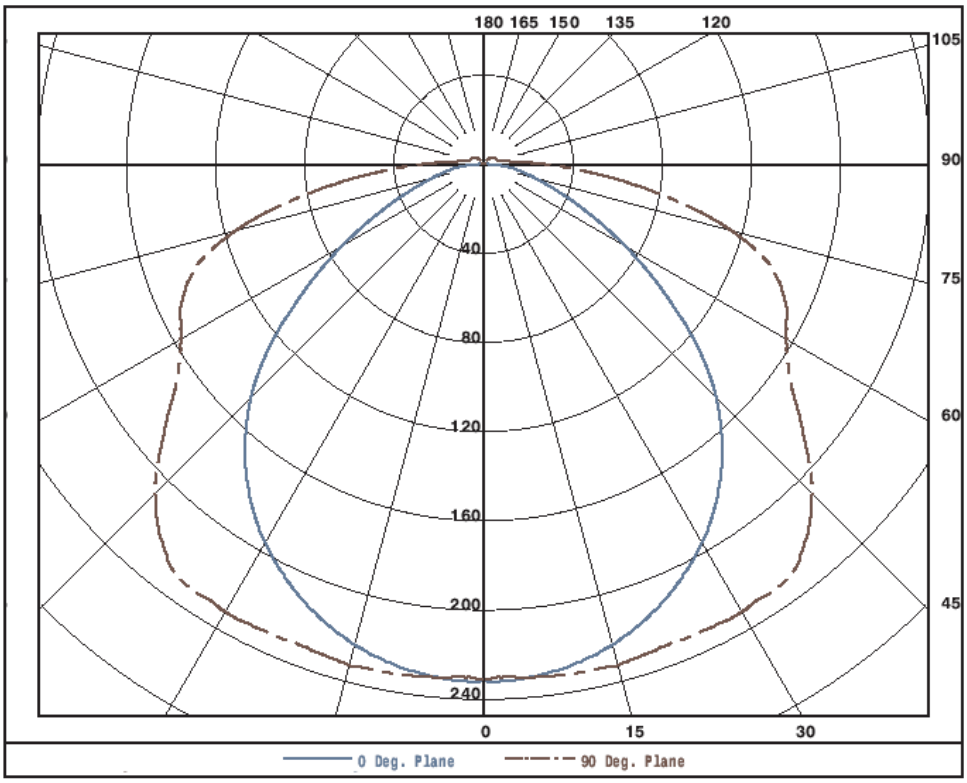


FIGURE 8. Candela distribution curves, prismatic window, symbols 347.2T8 and 347.3T8.

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TABLE I. Numerical candela distribution, white diffusing window, symbols 331.1T8 and 331.2T8.

Vertical angle	Candela distribution	
	0-degree	90-degree
0	228	228
5	227	227
15	216	223
25	195	211
35	166	193
45	132	172
55	98	150
65	64	125
75	32	95
85	7	66
90	2	55

Vertical angle	Candela distribution	
	0-degree	90-degree
105	1	46
115	0	35
125	0	26
135	0	19
145	0	13
155	0	8
165	0	4
175	0	0
180	0	0

TABLE II. Numerical candela distribution, prismatic window, symbols 347.2T8 and 347.3T8.

Vertical angle	Candela distribution	
	0-degree	90-degree
0	231	231
5	231	231
15	223	233
25	207	231
35	182	231
45	147	208
55	97	167
65	52	149
75	23	115
85	9	51
90	2	28

Vertical angle	Candela distribution	
	0-degree	90-degree
95	2	13
105	2	6
115	1	6
125	0	5
135	0	4
145	0	2
155	0	1
165		
175		
180		

PHOTOMETRIC DATA:

Photometry reports based on LM-41 and other pertinent IESNA procedures.

Plane: 0 degrees is parallel to lamp, 90 degrees is perpendicular to lamp.

Electrical values: 120 volts, 60 hertz, 18.0 watts input power.

Ambient temperature: 25±1 °C.

Lamp: GE F17T8-SP41-ECO (seasoned for at least 100 hours).

Lamp correction: lamp rated at 1325 lumens.

Ballast: furnished with fixture, see MIL-DTL-16377/81 for specification.

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REQUIREMENTS:

CLASSIFICATION:

Type: I (fluorescent).

Class: 2 (general).

SYMBOL NUMBERS:

331.1T8: Complete lighting fixture assembly, standard mounting, with one 17-watt T8 cool white fluorescent lamp and white diffusing window assembly.

331.2T8: Complete lighting fixture assembly, flush mounting, with one 17-watt T8 cool white fluorescent lamp and white diffusing window assembly.

347.2T8: Complete lighting fixture assembly, standard mounting, with one 17-watt T8 cool white fluorescent lamp, clear and white prismatic window assembly.

347.3T8: Complete lighting fixture assembly, flush mounting, with one 17-watt T8 cool white fluorescent lamp, clear and white prismatic window assembly.

DESIGN:

Electrical rating:	17 watts, 120 volts, 60 hertz, ungrounded electrical distribution system total input power: 18.0 watts at 120 volts.
Dimension and configurations:	See figures 1, 2, and 3.
Photometric data:	See figures 7 and 8 and tables I and II.
Wiring diagram:	See figure 6.
Ballast:	See MIL-DTL-16377/81 for specification.
Mounting dimensions:	
Symbols 331.1T8 and 347.2T8:	Standard mounting (surface), see figure 5.
Symbols 331.2T8 and 347.3T8:	Flush mounting (recessed), see figure 4.
Window assembly:	
Symbols 331.1T8 and 331.2T8:	White window, see table III and figure 27 of MIL-DTL-16377.
Symbols 347.2T8 and 347.3T8:	Prismatic window, see table III and figure 27 of MIL-DTL-16377.
Enclosure:	Watertight; see 3.5.11 of MIL-DTL-16377.
Cable:	10 feet (minimum) of type LSDHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbols 331.1T8 and 347.2T8:	Not to exceed 8 pounds, 10 ounces.
Symbols 331.2T8 and 347.3T8:	Not to exceed 10 pounds, 2 ounces.

DESIGNATION AND MARKING:

The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3:

a. Two Type I labels, equipment identification, reference MIL-DTL-16377 figure 11, shall include manufacturer and address, one located on back and other located on an inner nonreflecting surface.

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b. One Type II label, circuit wiring information, reference MIL-DTL-16377 figure 12, located on outside surface adjacent to cable entrance.

c. One Type IV label, repair parts, reference MIL-DTL-16377 figure 14, information to be shown shall include “USE THE FOLLOWING REPAIR PARTS ONLY.” List all standard parts shown in table III, located on an inner nonreflecting surface.

EXAMINATIONS AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article testing/inspection:

Complete fixtures:	See table IV.
Window assemblies:	See table V.
Comparison inspection:	See tables IV and V.
Conformance inspection:	See MIL-DTL-16377.

Inspection of product for delivery:

Examination:	See MIL-DTL-16377.
Conformance test:	See MIL-DTL-16377.
Examination for preparation of delivery:	See MIL-DTL-16377.

GENERAL INFORMATION:

Table III. Fixture standard parts.

Table IV. Complete fixture first article tests.

Table V. Window assembly first article tests.

Table VI. Complete fixture part numbers.

TABLE III. Standard parts.

Name	Military p/n or specification	National Stock Number (NSN)	Remarks
Lamp	Commercial Identification F17T8/CW	6240-01-391-6027	---
Lamplock	M16377/40-003	---	---
Lampholder	UL 542	6250-00-284-0481	Commercial
Ballast	M16377/81-001	---	Integrated starter type
Shockmount	M16377/41-008	6210-01-393-4302	0.067 steel, zinc plated
Window assembly, diffusing	M16377/8-001	6210-00-548-0601	1/
Window assembly, prismatic	M16377/8-003	---	1/
Filter, 17-watt T8 fluorescent lamp (DARK ADAPTATION)	M16377/42-010	---	Red transparent

NOTE:
1/ Window assemblies (diffusing and prismatic) are the same as T12 lamp fixtures (MIL-DTL-16377/8).

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TABLE IV. First article tests and comparison inspection for complete fixtures.[1/]

Test	Requirement	Test method	Remarks
Examination	3.13	4.5	---
Operation	3.13	4.8.1	---
Electromagnetic interference	3.6.14	4.8.16	---
Electrical equipment interface testing requirements: MIL-STD-1399-300	MIL-STD-1399-300	MIL-STD-1399-300	[2/], [3/]
Dielectric withstanding voltage	3.6.15	4.8.2	[4/]
Insulation resistance	3.6.16	4.8.3	[4/]
Enclosure effectiveness	3.5.11	4.8.14.4	[4/]
Shock	3.5.12	4.8.7	[4/]
Vibration	3.5.13	4.8.8	[4/]
Enclosure effectiveness	3.5.11	4.8.14.4	[4/]
Dielectric withstanding voltage	3.6.15	4.8.2	[4/]
Insulation resistance	3.6.16	4.8.3	[4/]
Noise	3.5.14	4.8.9	---
Continuity of grounding	3.6.13.4	4.8.12	---
Light output	3.7.1	4.8.6.1	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---

NOTES:

1/ Requirements and test methods use MIL-DTL-16377.

2/ If less, then 1KVA exception not applicable for input waveform THD.

3/ Solid state components shall be subject to 250 positive spikes and 250 negative spikes (500 total spikes), one spike per minute, using testing method identified in MIL-STD-1399-300.

4/ Tests shall be performed in the order specified.

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TABLE V. First article tests and comparison inspection for window assemblies. [1/]

Test	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Shock	3.5.12	4.8.7	[2/]
Vibration	3.5.13	4.8.8	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/]
Light output	3.7.1	4.8.6.1	Candela distribution

NOTES:
[1/] Testing requirements and methods use MIL-DTL-16377.
[2/] Tests shall be performed in the order specified.

TABLE VI. Complete fixture part numbers.

Name	Military p/n
Complete fixture, symbol 331.1T8	M16377/78-331.1T8
Complete fixture, symbol 331.2T8	M16377/78-331.2T8
Complete fixture, symbol 347.2T8	M16377/78-347.2T8
Complete fixture, symbol 347.3T8	M16377/78-347.3T8

Custodians:
Army – CR4
Navy – SH
Air Force – 99

Preparing Activity:
Navy – SH
(Project 6210-2006-003)

Review Activities:
Navy – AS, CG
Air Force – 03
DLA – GS2
GSA – FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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